AKTU paper | www.aktunotes.in

Printed Pages : 4

NAS401

(Following Paper ID and Roll No. to be filled in your Answer Book)

PAPER ID : **199419**

Roll No.

B. Tech.

(SEM. IV) THEORY EXAMINATION, 2014-15

ENGG MATHEMATICS - III

Time : **3** Hours] [Total Marks : **100**

Note: Attempt all questions from each Section as indicated. The symbols have their usual meaning.

1 Attempt any FOUR parts of the following: (5×4=20)

(a) Using C – R equations show that $f(z) = |z|^2$ is not analytical at any point.

(b) State Cauchy's integral formula. Hence evaluate:

$$\int_c \frac{2z+1}{z^2+z}\,dz \qquad \text{where C is } |z| = \frac{1}{2}.$$

(c) Expand $\frac{1}{Z^2-3Z+2}$ in the region $1 < |z| < 2$.

(d) Find the analytic function whose real part is $e^{2x}(x\cos 2y - y\sin 2y)$

Source: AKTU official examination paper - www.aktunotes.in

AKTU paper | www.aktunotes.in

(e) Evaluate the integral:

$$\int_0^{2\pi} \frac{d\theta}{5-3\cos\theta}$$

(f) Determine the poles of the following function and residue at each pole:

$$f(z) = \frac{z^2}{(z-1)^2(z+2)}$$

and hence evaluate

$$\int_c f(z)\,dz, \quad \text{where } C: |z| = 3.$$

2 Attempt any TWO parts of the following: (10×2=20)

(a) Find the Fourier cosine transform of $\frac{1}{1+x^2}$ and hence find Fourier sine transform of $\frac{x}{1+x^2}$.

(b) Find the inverse Z-transform of:

$$f(z) = \frac{3z^2 - 18z + 26}{(z-2)(z-3)(z-4)}$$

(c) Solve the following difference equation:

$$y_{k+2} + 4y_{k+1} + 3y_k = 3^k; \text{ given } y_0 = 0 \text{ and } y_1 = 1.$$

AKTU paper | www.aktunotes.in

3 Attempt any TWO parts of the following: (10×2=20)

(a) Using the method of least square fit a curve of the form $y = a\,b^x$ to the following data:

X	2	3	4	5	6
Y	8.3	15.4	33.1	65.2	127.4

(b) Find the mean and variance of Binomial distribution.

(c) The first four moments of a distribution about x = 4 are 1, 4, 10 and 45. Calculate the moments about the mean and comment upon the Skewness and Kurtosis of the distribution.

4 Attempt any TWO parts of the following: (10×2=20)

(a) Using Newton-Raphson method, find real root of equation $3x - \cos x - 1 = 0$ correct up to four decimal places. Also find the order of convengence of the method.

(b) From the given data

X	4	5	7	10	11	13
$f(x)$	48	100	294	900	1210	2028

find the polynomial in x and hence find the value of $f(8)$.

(c) Estimate from the following table the number of students who obtained marks between 40 and 45:

Marks	30 – 40	40 – 50	50 – 60	60 – 70	70 – 80
No. of Students	31	42	51	35	31

AKTU paper | www.aktunotes.in

5 Attempt any TWO parts of the following: (10×2=20)

(a) Apply Gauss - Seidal Method to solve the equations (three iterations) :

$2x+10y+z = 55$

$10x+y+2z = 44$

$x+2y+10z = 61$

(b) Find $\int_0^6 \frac{e^x}{1+x}\,dx$ approximately using Simpson's $\frac{1}{3}$ rule and Simpson's $\frac{3}{8}$ rule, dividing the range into 6 equal parts.

(c) Given the initial value problem $\frac{dy}{dx} = 1 + y^2; \quad y(0) = 0.$ Find $y(0.2),\ y(0.4)$ by Runge - Kutta fourth - order.